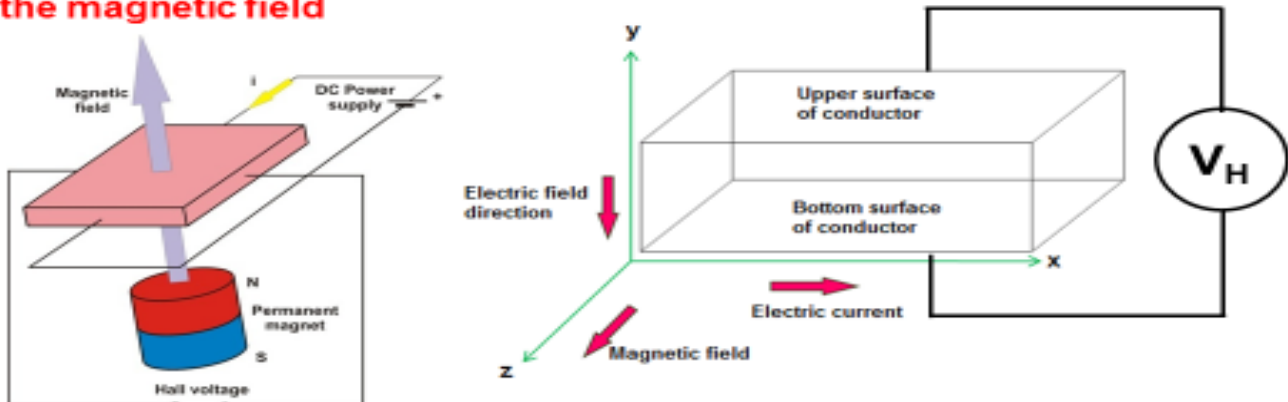


22PH202 – ELECTROMAGNETISM AND MODERN PHYSICS

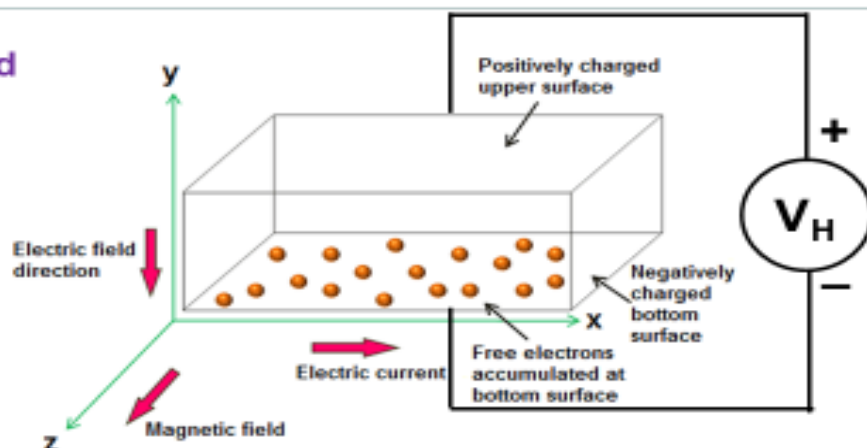
Unit-V: ENERGY BANDS IN SOLIDS_LM6

Hall Effect (Discovered by Edwin Hall in 1879)

When a magnetic field is applied to a current carrying conductor or semiconductor in a direction perpendicular to the flow of current, an electric field (Hall Voltage) will be created inside the conductor or semiconductor in a direction perpendicular to both the current and the magnetic field



Magnetic field is applied in a direction perpendicular to the flow of current (z-direction)



Hall Effect in conductor

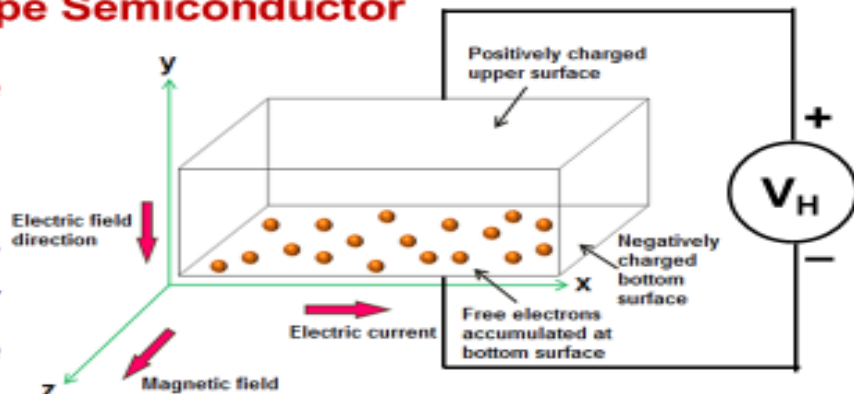
The electric field produced will exerts a force in the negative y-direction (downwards)

The electric field will pushes the electrons downwards (in the negative y-direction) (Hall voltage will be negative – due to electrons)

Hall Effect in n-type Semiconductor

Consider an **n-type** semiconductor

A magnetic field is applied perpendicular to the direction of the current



Hall Effect in N-type semiconductor

Both **free electrons and holes** are pushed down towards the **bottom surface** of the n-type semiconductor

Holes are **minority carriers** in n-type (negligible)

More number of electrons are accumulated at the **bottom surface**

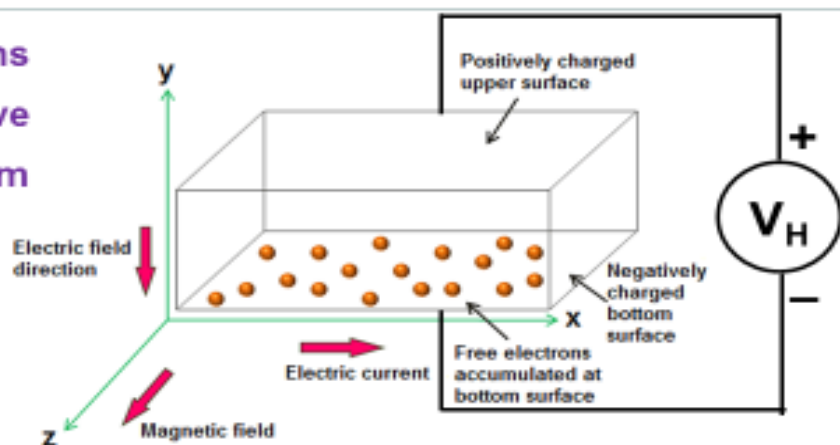
Accumulated electrons produces more negative charge on the bottom surface

Bottom surface

Negatively charged

Upper surface

Positively charged



Hall Effect in N-type semiconductor

A potential difference (voltage) is developed between the upper and bottom surfaces in the **negative -y-direction**

The voltage developed (**Hall voltage**) will be **negative**

Due to the applied magnetic field, the electrons move towards the bottom surface with a velocity ' v '

The potential difference created between the upper and bottom surface will give rise to an electric field in the negative y-direction

Force due to the potential difference $= -e E_H$

Force due to the magnetic field $= -B e v$

At equilibrium condition

$$\begin{aligned} -e E_H &= -B e v \\ E_H &= B v \end{aligned}$$

Current density (J_x) in the x-direction

$$J_x = -n_e e v \Rightarrow v = -\frac{J_x}{n_e e}$$

$$E_H = B v$$

$$v = -\frac{J_x}{n_e e}$$

$$E_H = B \left(-\frac{J_x}{n_e e} \right) = -\frac{J_x B}{n_e e} = -\frac{1}{n_e e} J_x B$$

$$E_H = -\frac{1}{n_e e} J_x B \Rightarrow E_H = R_H J_x B \quad (1)$$

$$R_H = -\frac{1}{n_e e}$$

R_H – Hall Coefficient

Negative sign indicates that the semiconductor is n-type

Similarly we can derive Hall coefficient for an P Type semiconductor

$$R_H = \frac{1}{n_h e}$$

R_H – Hall Coefficient

Positive sign indicates that the semiconductor is p-type

Hall Coefficient in terms of Hall Voltage

$$E_H = R_H J_x B$$

$$R_H = E_H / J_x B$$

If $J_x = 1 \text{ amp/m}^2$ and $B = 1 \text{ Tesla}$

$$R_H = E_H$$

Half coefficient (R_H) is defined as the Hall field developed per unit current density per unit applied magnetic field.

If the thickness of the sample is 't' and the voltage developed is ' V_H ' then Hall voltage

$$V_H = E_H t \quad \text{————— (2)}$$

Substituting eqn. (1) in eqn (2) we have

$$V_H = R_H J_x B t \quad \text{————— (3)}$$

We know, current density $J_x = I_x / A$

Area of the specimen $A = bt$

b - is the breadth of the sample ; t - is the thickness of the sample

$$\text{Then, current density } J_x = I_x / bt \quad \text{————— (4)}$$

Substituting eqn. (4) in eqn (3) we get

$$V_H = R_H I_x B t / bt$$

$$R_H = V_H b / I_x B$$

R_H - Hall Coefficient

V_H - Hall Voltage

b- Breadth of the sample

I_x = Current through the sample

B= Magnetic field applied

Application of Hall Effect

1. The sign (N-type (or) P-type) of charge carriers can be determined.
(N-type = R_H is negative; p-type = R_H is positive)
2. The carrier concentration can be determined $\left[n = \frac{1.18}{qR_H} \right]$
3. The mobility of charge carriers is measured directly $\left[\mu = \frac{-\sigma R_H}{1.18} \right]$.
4. Electrical conductivity can be determined $[\sigma = n q \mu]$.
5. It can be used to determine whether the given material is metal, insulator, or semiconductor and the type of the semiconductor.
6. It can be used to determine the power flow in an electromagnetic wave.